Slides are available on my website.
`http://lukaseigentler.github.io`

Variation in resource quality causes maintenance of individual variation in a competitive trait

Behavior, Evolution & Emergence (BEE'26)

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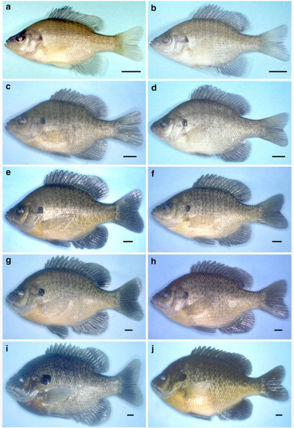
Lukas Eigentler (University of Warwick)
joint work with
Klaus Reinhold (Bielefeld), David W. Kikuchi (Oregon State), Valeria Giunta (Swansea)

Individual variation

- Evolution does not necessarily lead to identical individuals.
- Individual variation is ubiquitous in many populations.
- This includes genetic variation.
- Potential causes: temporal/spatial environmental variability, negative frequency-dependent selection, ...



Bluegill size differences[1]

[1] (Yokogawa, K.: *Ichthyological Research* 60.1 [2012])

Individual variation in competitive traits

- Traits determining intraspecific competitiveness often vary within populations.
- Examples: armaments, body size, sensory capabilities to detect food, ...
- Investment into competitiveness typically bears a cost.
- Trade off between ability to attain resources and resource investment into reproduction.

Source: Wikimedia Commons

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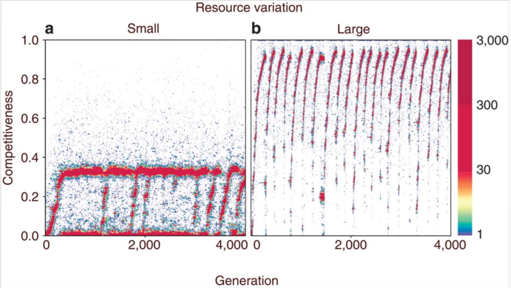

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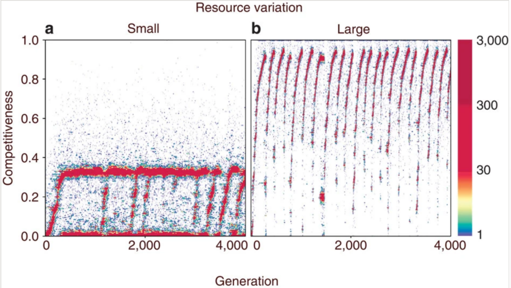
- What if resource qualities also vary continuously?



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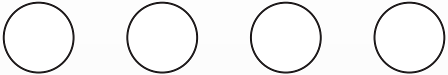
Individual based model



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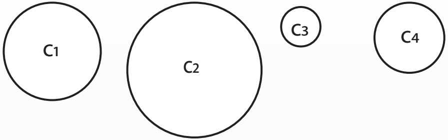

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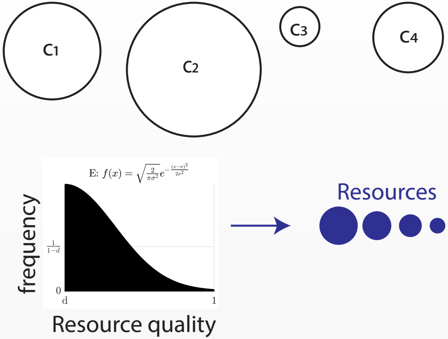

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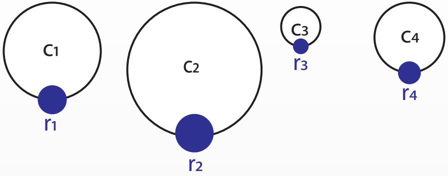

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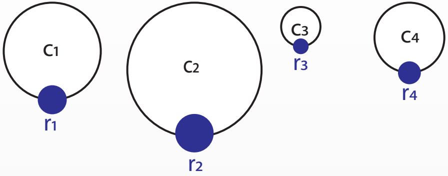

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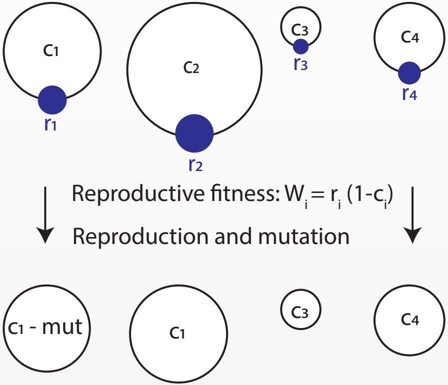
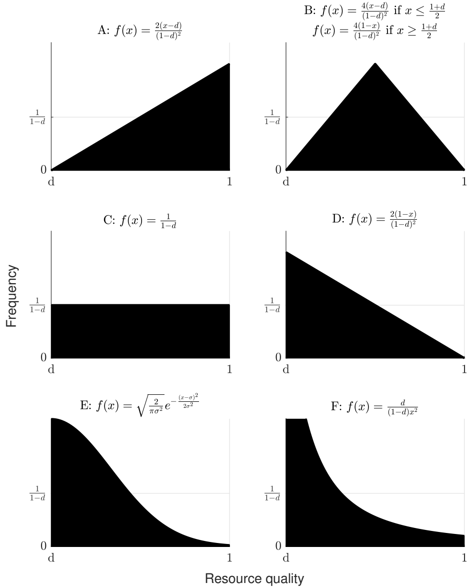
- Next generation is created through draw with replacements with weights W_i and mutations.



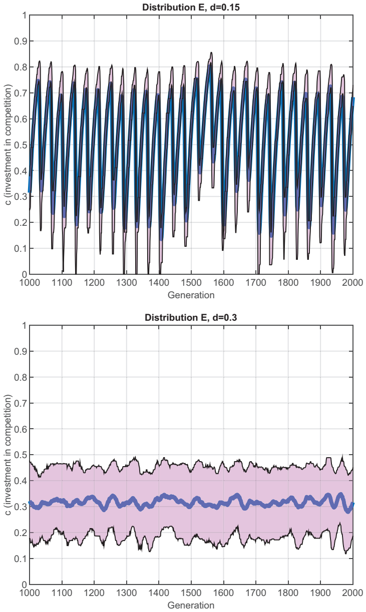
Resource distributions

- A: linearly increasing
- B: linearly increasing and decreasing
- C: uniform
- D: linearly decreasing
- E: half-normal
- F: power law



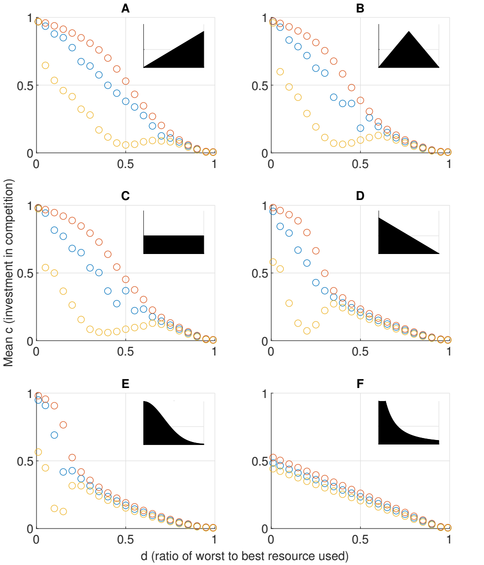
Arms races vs. stable polymorphisms

- Dynamics either represent arms races or stable polymorphisms.
- Arms race: mean competitive trait oscillates with little individual variation at a time.
  If all individuals have $c = c^*$, then $c = c^* + \varepsilon > c^*$ has higher fitness.
  If c^* is high, low competitive traits can invade; in particular, $c_1 = 0$ (fitness $W = (1 - c_1)d = d$) has larger fitness than $c_2 > 1 - d$ because $W = (1 - c_2)r < dr \leq d$.
- Stable polymorphism: mean competitive trait remains constant over time; large individual variation at all times.



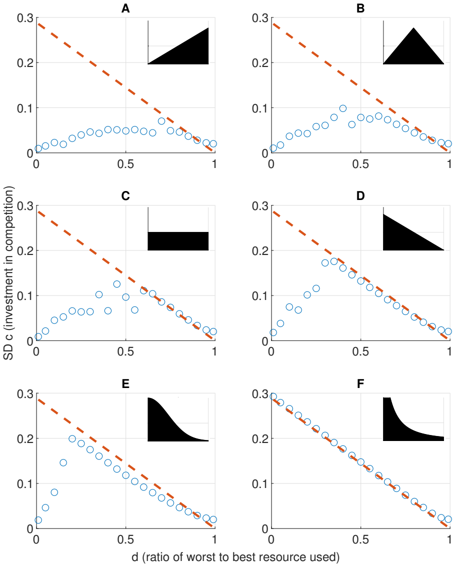
Worst-to-best resource ratio

- The ratio d of the worst to the best resource determines mean investment into competition.
- Arms races occur when d is small.
- Period of arms race cycles increases as d decreases; unclear behaviour for $d \to 0$.
- No arms races for power law distribution.
- Stable mean trait values occur when d is large.
- The threshold $d = d^*$ depends on the resource distribution.



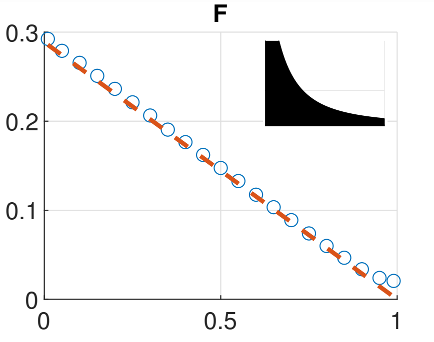
Worst-to-best resource ratio

- Little individual variation occurs during arms races.
- Stable mean trait values are always associated with stable polymorphisms.
- Individual variation is close to maximum (uniform distribution with $0 \leq c \leq 1 - d$).



Special case: power law

- Can show analytically that the power law with pdf $g(r) = \frac{d}{(1-d)r^2}$ leads to uniform trait distribution.



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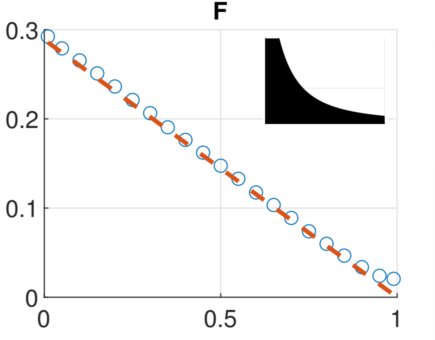

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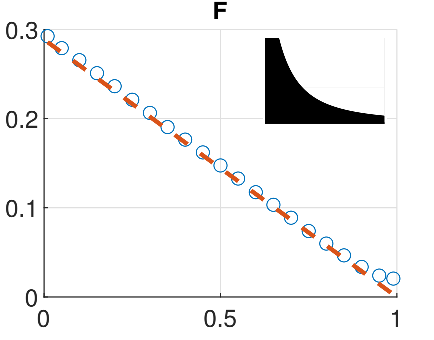

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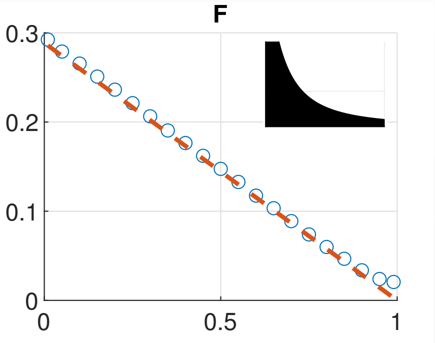
- Hierarchical resource assignment means $P(r < r^*) = P(c < c^*)$, where $r^* = \frac{d}{1-c^*}$. Therefore, the cdf of the trait distribution is

$$P(c < c^*) = P(r < r^*) = \frac{r^* - d}{(1-d)r^*} = \frac{c^*}{1-d}.$$

$\Rightarrow$ uniform distribution.

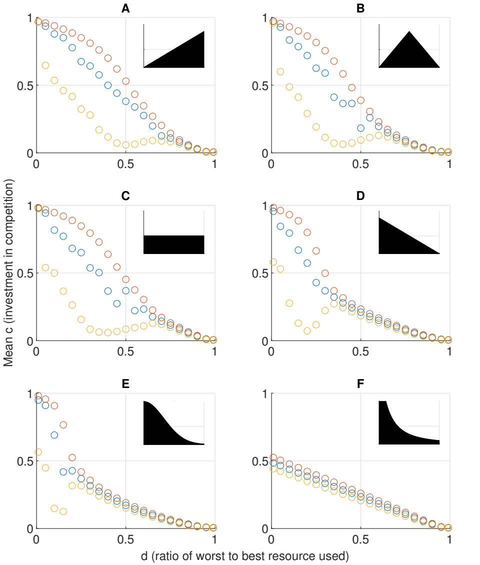


Conclusions – IBM

- Continuous individual variation in a competitive trait is a consequence of intraspecific competition for resources with continuously varying quality.
- The ratio between worst and best quality resources determines (i) mean investment into competition, and (ii) whether arms races with little individual variation or stable polymorphisms occur.
- The distribution of resource quality is important.
- Resource distributions with more low-quality resources lead to stable polymorphisms in more cases.

What next? Nonlocal PDE!

- Main limitation: IBMs are hard to analyse.
- PDE models allow for bifurcation analysis.
- Formal derivation of a PDE from the IBM is (we think) not possible.



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- Mutation: mutation in a single gene will have very small impact on trait value $\Rightarrow$ diffusion.
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- $$\frac{\partial u(t,c)}{\partial t} = \frac{\partial^2 u(t,c)}{\partial c^2} + \lambda(t,c)u(t,c).$$

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- Fitness of phenotype c

$$W(t,c) = \underbrace{(1-c)}_{\text{proportion left for reproduction}} \cdot \underbrace{r(t,c)}_{\text{resource quality attained}}.$$

Resource quality

- For $0 \leq d < 1$, let the pdf of the resource distribution be $g : [d, 1] \to [0, \infty)$, such that the inverse $G^{-1} : [0, 1] \to [d, 1]$ of its cdf $G : [d, 1] \to [0, 1]$ exists.

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-

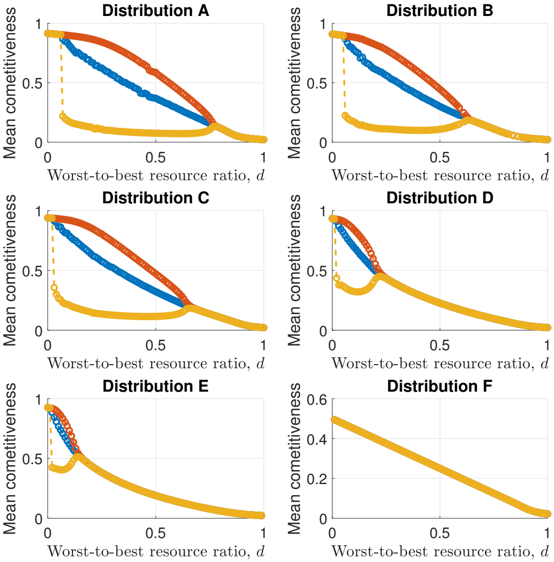
$$\Rightarrow r(t, c) = G^{-1}(U(t, c)).$$

Full nonlocal model

$$\frac{\partial u(t,c)}{\partial t} = \frac{\partial^2 u(t,c)}{\partial c^2} + \left[(1-c)G^{-1}\left(\int_0^c u(t,\tilde{c})\mathrm{d}\tilde{c}\right) - \int_0^\infty (1-c)G^{-1}\left(\int_0^c u(t,\tilde{c})\mathrm{d}\tilde{c}\right) u(t,c)\mathrm{d}c\right] u(t,c).$$

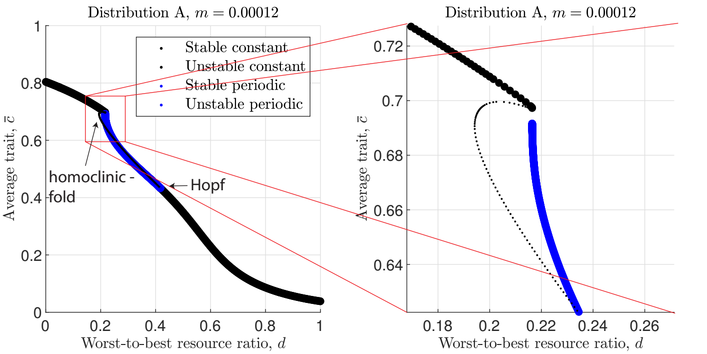
Numerical simulations

- Results qualitatively the same as for IBM.
- Note lack of oscillations for small d.



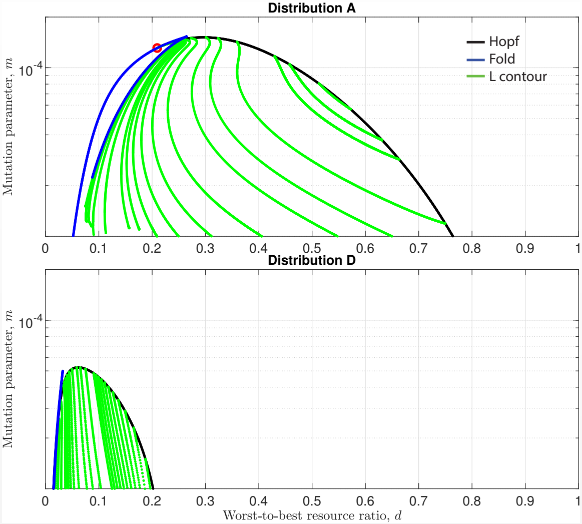
1d numerical bifurcation analysis **work in progress**

Tool: discretise trait space $\Rightarrow$ continuation of large ODE system using AUTO-07p.



From high d to low d: constant sol $\rightarrow$ onset of periodic sol via Hopf bifurcation $\rightarrow$ termination of periodic sol at either Hopf bifurcation or homoclinic-fold bifurcation $\rightarrow$ constant sol.

2d numerical bifurcation analysis: *m*-*d* **work in progress**



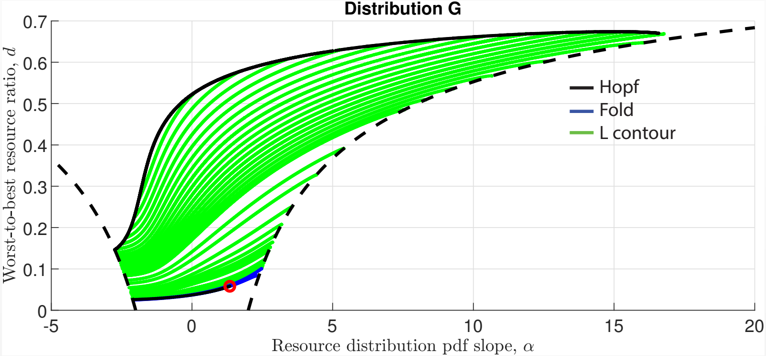
Periodic sol require sufficiently low mutation rate. Onset at low d occurs at Hopf bifurcation for "large" m; at homoclinic-fold bifurcation at "small" m.

2d bif analysis: generalised linear distribution **work in progress**

Distributions A, D are special cases of general distribution with pdf

$$g(r;\alpha) = \alpha r + \beta, \quad d \leq r \leq 1, \quad -\frac{2}{(1-d)^2} \leq \alpha \leq \frac{2}{(1-d)^2}$$

$\Rightarrow$ Continuation in α to study dynamics for this generalised distribution.



Analytical results **work in progress** – Valeria Giunta

We can show analytically that

- the existence of lower and upper bounds for steady state solutions.
- if the fitness is monotonically decreasing with the trait, the trait distributions are monomodal with peak at $c = 0$.
- $d > \overline{d}$ is a sufficient condition for constant solutions to be stable.

Conclusions – PDE

- Numerical bifurcation analysis of PDE model is possible.
- Pattern onset at Hopf bifurcation (high/low d) and homoclinic-fold bifurcation (low d).
- Can perform 2-parameter continuation of these bifurcations.
- Low mutation rates required for periodic "arms race" solutions.
- Can find codimension 2 bifurcation, such as Bogdanov-Takens.
- Can perform continuation in a parameter describing the resource quality distribution to understand how different distributions affect results.

Open questions

- What are the most realistic resource quality distributions?
- How to include more realism, such as spatial dynamics or sexual reproduction?
- Are there better tools for numerical bifurcation analysis of nonlocal PDEs (BifurcationKit, pde2path)?
- Can we formally derive a continuum model from the IBM?

References

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[1] Reinhold, K., Eigentler, L. and Kikuchi, D. W.: 'Evolution of individual variation in a competitive trait: a theoretical analysis'. *J. Evolution. Biol.* 37.5 (2024), pp. 538–547.